Product Pricing:
Maximize Lyft’s net revenue in a new city

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Case: Maximize Lyft's net revenue on airport route in Toledo

Problem Statement

- Pretend you're the pricing product manager for Lyft's ride-scheduling feature, and you're launching a new city like Toledo, Ohio
- The prevailing rate that people are used to paying for rides from the airport to downtown (either direction, one way) is $25. The prevailing wage that drivers are used to earning for this trip is $19
- You launch with exactly this price: $25 per ride charged to the rider, $19 per ride paid to the driver. It turns out only 60 or so of every 100 rides requested are finding a driver at this price
- Your goal is to maximize net revenue for the next 12 months on this route in Toledo

Base Assumptions:

- Prevailing rate charged to the riders cannot be increased
- Net revenue = Amount riders pay - Amount Lyft pays out to drivers
- At the prevailing wage, drivers have a 5% monthly churn rate and complete 100 rides / month. Churn and ride completion would vary for different wages
- CAC for both riders & drivers is sensitive to the corresponding rate of acquisition
- While there is more than one route to think about in Toledo, for the sake of this exercise you can focus on this one route

Understanding Pricing Dynamics & Unit Economics

Pricing

Current Scenario		
Prevailing rider rate	$25	per ride
Prevailing driver Wage	$19	per ride
Lyft Margin	$6	per ride
Booking success rate	60%	

Pricing Experiment		
Rider rate	$25	per ride
Driver Wage	$22	per ride
Lyft Margin	$3	per ride
Booking success rate	93%	

Assumption:
Lyft starts serving the market with 3000 rides per month

Unit Economics

Driver-side Unit Economics		
Min CAC	$400	per user
Max CAC	$600	per user
Monthly rides completed	100	per month
Monthly Churn	5%	per month

Rider-side Unit Economics		
Min CAC	$10	per user
Max CAC	$20	per user
Monthly rides requested	1	per month
Trigger event	Failed to find a rider	
Churn for users NOT experiencing trigger event	10%	per month
Churn for users experiencing trigger event	33%	per month

Iterative approach: Consider scenarios

Maximizing Net Revenue when there are no check metrics

Scenario 1: Assumptions

- Market is constrained, ie in the coming 12 months, Lyft can serve maximum 3000 rides per month
- The company's goal is to focus only on Net revenue without any check metrics

		NSM
Lyft Margin per ride	**Success Rate**	**Annual Net Revenue**
$6	60%	$129,600
$5	71%	$127,800
$4	82%	$118,080
$3	93%	$100,440

Click here for calculations

- As a Pricing Product Manager with no check metrics, my only goal would be to maximise Net Revenue
- In such a scenario, I'll proceed with charging $6 as Lyft's margin and **paying $19 per completed ride to the drivers**

Numbers in red are hypothesised

Scenarios with check metrics

- In real-world scenarios, for a late stage startup, it makes more sense for a company to minimise burn and focus on profitability
- Further scenarios will try to analyse possibilities under such conditions

Assumptions:

- As CAC for rider & driver change with corresponding rates of acquisition, consider the following tables
- With higher wage available, drivers are inclined to churn less and complete more rides in a month

Rider CAC corresponding to churn rate		
Churn%	Acquisition rate	CAC
10.0%	>10%	$10
15.0%	>15%	$14
20.0%	>20%	$17
25.0%	>25%	$20

Driver CAC, Churn & Rides completion corresponding to Driver Wage				
Driver Wage	Rides per month	Churn%	Acquisition rate	CAC
$19	100	5.0%	5.0%	$600
$20	110	4.8%	4.8%	$550
$21	120	4.6%	4.6%	$480
$22	130	4.4%	4.4%	$400

Numbers in red are hypothesised

Scenario 2 with check metrics in a constrained market

Scenario 2: Assumptions

- Market is constrained, ie in the coming 12 months, Lyft can serve maximum 3000 rides per month
- Along with maximising Net revenue the company also wants to optimise for Contribution Margin

		NSM	Check Metrics		
Lyft Margin	Success Rate	Annual Net Revenue	Rider CAC	Driver CAC	Contribution Margin
$6	60%	$129,600	$166,604	$10,800	-$47,804
$5	71%	$127,800	$129,885	$8,640	-$10,725
$4	82%	$118,080	$90,308	$6,624	$21,148
$3	93%	$100,440	$49,267	$4,874	$46,299

Click here for calculations

- In a constrained market, my priority would be to maximise Net revenue while optimising for the contribution margin
- In such a scenario, I'll proceed with charging $4 as Lyft's margin and **paying $21 per completed ride to the drivers**

Numbers in red are hypothesised

Scenario 3 with check metrics in a growing market

Scenario 3: Assumptions

- Market is growing ie Rides available are growing at the rate of 5% month-on-month
- Along with maximising Net revenue the company also wants to optimise for Contribution Margin

		NSM	Check Metrics		
Lyft Margin	Success Rate	Annual Net Revenue	Rider CAC	Driver CAC	Contribution Margin
$6	60%	$172,022	$233,283	$14,335	-$75,596
$5	71%	$169,633	$183,601	$9,815	-$23,783
$4	82%	$156,732	$151,315	$8,158	-$2,741
$3	93%	$133,317	$74,844	$6,469	$52,004

Click here for calculations

- In a growing market, my priority would be to maximise Net revenue while minimising losses
- In such a scenario, I'll proceed with charging $3 as Lyft's margin and **paying $22 per completed ride to the drivers**
- This is also because for 24-month calculation period for the above table, all other options have a negative contribution margin due to high rider churn

Numbers in red are hypothesised

Summarising the solution

Scenario	Lyft Margin	Driver Wage	Annual Net Revenue
No check metrics	$6	$19	$129,600
Check metrics & constrained market	$4	$21	$118,080
Check metrics & growing market	$3	$22	$133,317

Other possibilities to deliberate

- Optimising for driver recruitment based on actual ride acceptance rate
- Calculating rides per month & growth in rides per month based on seasonal trends
- Experimenting with changing Lyft margin based on seasonal trends

Numbers in red are hypothesised